CALVIN GALAGAIN

Data Scientist & A.I.

Engineer & Researcher

Looking for an PhD offer

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PROFESSIONAL EXPERIENCE

Detection and tracking of objects in real time on LiDAR data

Artificial Intelligence RESEARCH - Exwayz (Startup - Station F)

01 April 2023 – 30 September 2023 Paris, 77013 France

- Implementation of a method to segment a 3D LiDAR frame and track all objects in real time (C++, Python).
- SLAM algorithm and SLAM-GPS fusion.
- Management of the anisotropy of the input data and efficient implementation of segmentation without degrading the accuracy of the method.
- 3D tracking of a multitude of objects with different dynamics, estimation of their state and bounding boxes of objects.

AI research

Zentrum für Recht und Digitalisierung - ZRD Saarbrücken

01 May 2022 – 30 September 2022 Saarbrücken, 66123 Deutschland

- Internship in a Research center which allowed me to become aware of the legal issues behind the algorithms we use
- Detection of sensitive content in a video (in partnership with the Prosecutor)
- Creation of a real-time and automatic database (**Python** : involving detection and tracking of the subject)
- Learned the methods to "hack" a neural network and extract information from the models (**Python** : Scraping, Adversarial Learning, GAN, Neural Network bruteforcing)

Reconstruction and control of faces by Deepfakes

Data Science & Artificial Intelligence RESEARCH - IRCGN

01 November 2021 – 30 September 2022 Paris, 93160 France

- Estimated the facial resemblance between a suspect and a person on a police video
- Creation of usable data by following a researcher's specifications
- Worked on adapted deep learning models (**Python** : Data Augmentation, Face Recognition, High performance Face Landmark Detection, Deepfakes Deep Learning Decoder from Scratch)
- Discovering the world of research
- Be critical of the work of other researchers

HOBBIES

- French equestrian championship (15 years), to compare myself to others
- Piano (classical) (14 years), to improve my creativity
- Chess, to relax while thinking
- Passionate about Entomology and nature, particularly Myrmecology (study of ants), to keep my feet on the ground

EDUCATION

01/10/2022 - 30/04/2023 M2 MVA Mathématiques / Vision / Apprentissage
ENS Paris Saclay - 91190 Gif-sur-Yvette (France)

01/09/2020 - 31/07/2023 Data Science & Artificial Intelligence Student (valedictorian)
Five-year program leading to the equivalent of an MSc in Engineering
ESIEE Paris - 93160 Noisy-le-Grand (France)

01/09/2018 - 31/07/2020 A two-year post-A level course in Maths and Physics to go directly into the 3rd year at engineering school
SAINT STANISLAS, 44000 Nantes (France)

20/06/2018 Baccalauréat S
Equivalent to English A-Levels in pure Maths and Physics (awarded with distinction)

HARD SKILLS

- **Python** (Keras, Tensorflow, OpenCV, NumPy, Pygame, Scikit-learn, PyTorch, Pandas, Seaborn, Statsmodels, …)
- R, Java, C, C++, Matlab, SQL, Caml, LaTeX
- Embedded computing (Arduino, Raspberry, ESP)
- French native
- English (B2) / TOEIC : 850
- German (B1)

SOFT SKILLS

- Adaptability
- Rigor
- Teamwork
- Ability to learn and discover
- Interpersonal skills

References available on request

PREVIOUS - PROFESSIONAL EXPERIENCE

Sentiment Analysis

Data Science & Artificial Intelligence Project – ESIEE

01 May 2021 – 30 September 2021 Paris, 93160 France

- End of year project centered on Keras with team management and specific knowledge in order to provide innovative and sufficiently evolved work to be usable
- Conformed to strict deadlines to write weekly technical points and thus monitor the progress of this work
- Processing and analysis of complex images to understand feelings on a live face which involves understanding neural networks and emotional characteristics of the face (**Python** : Data Augmentation, Transfer learning, Fine Tuning, DCNN)
- Text processing and analysis to understand the sentiments that are expressed using NLP (**Python** : word2vec, gensim, nltk, stopwords, text classification)

ESIEE Admissions officer

ESIEE Paris

01 June 2021 – 31 July 2021 Paris, 93160 France

- Recruited the most promising students for the school knowing how to manage a project with schedules, a budget, communication and meetings
- Good pitcher and good communicator to succeed in motivating and uniting students around a common test
- Demonstrated endurance and managed pressure

Science and Computer science Teacher

Evolukid

01 August 2020 – 30 September 2022 Paris, 93160 France

- Taught math, physics, Python, Matlab and Arduino for students of all levels and dedicated time to provide a solid education
- Empathy and pedagogy complimented by many voices that testify to multiple skills
- Listens to have continuous and personalized follow-up by being attentive and respectful to enable the student to succeed

Salesman and Exotic Pet Sitter

Reptiligne

01 June 2016 – 31 July 2016 Gazeran, 78125 France

- Looked after animals by applying rigor and caution for flawless health for customer requests
- Discovered the corporate world and the team spirit necessary for good communication

ESIEE PROJECT (details available on request)

Reinforcement Learning

ESIEE Paris

- **Python** : Tensorflow, Keras, sklear, Seaborn
- Predicting signals
- SimpleRNN, GRU, LSTM
- Neural Network from Scratch to predict MNIST
- Denoising EEG

3-state classification problem

ESIEE Paris

- **Python & R**
- Feature Engineering, Feature Selection
- Handling Imbalanced dataset
- SVM, Random Forest, Boosting (CatBoost, GradientBoosting, AdaBoost, XGB), KNeighbors, MLP
- bootstrapping and cross-validation

Data Visualization

ESIEE Paris

- **Python & R** : Dash, Flask, Mongo, Redis, BeautifulSoup, plotly, yahoo_fin
- Predicting stocks and market indicators
- Scraping Web
- Modeling climate change and real estate
- Working with Docker & Github

Semantic AI

ESIEE Paris

- Creating ontology
- Using Protégé to deal with intelligent systems
- Reflecting on the veracity of a source of knowledge

MVA PROJECT (details available on request)

Reinforcement Learning

MVA

Course & Project Content :

- Historical multi-disciplinary basis of reinforcement learning
- Markov decision processes and dynamic programming
- Stochastic approximation and Monte-Carlo methods
- Function approximation and statistical learning theory
- Approximate dynamic programming
- Introduction to stochastic and adversarial multi-arm bandit
- Learning rates and finite-sample analysis
- **Python**

Deep Learning

MVA

Course Content :

- History of deep learning and its relation to cognitive science
- Feedforward networks, regularisation and optimization
- Representation learning & Siamese networks
- GAN & transfer learning
- Recurrent networks and LSTM for Natural Language Processing
- Object detection

Project :

- Sentiment analysis on a face
- Transformers and CNN reprogrammed from scratch
- Visualization of Transformers attention
- **Python**

Point Clouds and 3D Modeling

MVA

Course Content :

- 3D Perception Systems
- Processing and Operators
- Registration
- Point Cloud Segmentation
- Curve and Surface Reconstruction
- Primitive-Based Modeling
- Rendering of Point Clouds and Meshes

Project :

- Deep learning for point cloud classification
- Data preprocessing and feature extraction
- Region segmentation
- Ground extraction
- Rasterization
- Relative height estimation
- **Python**

Object recognition and computer vision

MVA

Course Content :

- Instance-level recognition: camera geometry, local-invariant features, correspondence, efficient visual search
- Category-level recognition: bags of features, sparse coding and dictionary learning
- Neural networks, optimization methods
- Convolutional neural networks for visual recognition
- Motion and human actions
- 3D object recognition
- Weakly-supervised learning for visual recognition

Project :

- Sentiment analysis on a face
- Transformers and CNN reprogrammed from scratch
- Visualization of Transformers attention
- **Python**
- Instance-level recognition, Neural networks, Bird recognition

Introduction to digital imaging

MVA

Course Content :

- Image acquisition
- Introduction to Image Restoration
- Radiometry and image dynamics
- Colorimetry and color perception
- Detection and representation of geometric structures
- Representation and synthesis of textures and inpainting
- Statistics of natural images and modeling

Project :

- Texture interpolation (Inpainting, ADaIN, alpha-blending, brushing, masking)
- **Python**

Deep Learning in Practice

MVA

Course & TPs Content :

- Deep learning, classical machine learning, optimization.
- Interpretability, explainability, visualization, grad-CAM.
- Architectures, neural networks, CNNs, RNNs, Graph-NN.
- Small data, noisy data, weak supervision, transfer learning, priors.
- Modeling, deep learning, physics, invariances, priors, physical properties.
- GAN, VAE, Normalizing Flows, Auto-ML, Auto-DeepLearning.
- Guarantees, generalization, formal proofs.
- **Python**

Advanced learning for text and graph data ALTEGRAD

MVA

Course Content :

- Graph based Text Mining
- Word & doc embeddings
- Deep learning for NLP
- Graph kernels, community detection
- Deep Learning for Graphs - Node classification
- Deep Learning for Graphs - Graph classification
- Sets embeddings - point clouds
- Network Architecture Search - interpretability

Project :

- **Python**
- Pre-trained models for protein classification : DeepCov, ProtTrans, DeepSite, PSSM-CNN, and ProtHINT
- Classify a dataset of 6,111 proteins into 18 different classes based on their sequence and structure
- Analysis of sequence data and label distribution, identification of underrepresented amino acids and classes
- Implemented Graph Neural Network (GNN) with message passing layers to capture protein structure information and perform graph classification

Detection theory and its industrial applications

MVA

Course Content :

- Application areas of detection: geometric structures, industrial anomalies, satellite cloud detection, disruptive events, etc.
- A contrario detection theory
- Construction of suitable models for each detection problem.
- In-depth analysis of background data: sampling, calibration, and validity of stochastic variables.
- Detailed case studies to illustrate the taught concepts and introduction to unresolved problems.

Project :

- **Python**
- Detection Theory project for high-resolution satellite image analysis.
- Goal: Quickly assess damage after a disaster by avoiding false differences between images.
- SIFT and SAR methods used for interest point detection and matching.
- A contrario approach to identify unusual differences.
- Background model based on matched keypoints density.
- Calculation of NFA (Number of False Alarms) for change detection.
- Implementation of keypoints detection and matching algorithm.
- Limitations: Different image formats, low resolutions, distortion, cloudy backgrounds.
- Potential improvements: Automatic parameter adaptation, alternative feature representation methods.
- Future application: Image fusion for extended coverage.

Algorithms for speech and natural language processing

MVA

Course Content :

- Speech features & signal processing
- hidden markov & finite state modeling
- probabilistic parsing
- continuous embeddings
- deep learning for language-related tasks (DNNs, RNNs)
- linguistics and psycholinguistics
- comparing human and machine performance

Project :

- Deep learning models for Named Entity Recognition (NER)
- Domain adaptation and Fine-tuning of existing NER models
- Data enrichment for African languages
- Utilizing pre-trained models such as XLM-RoBERTa
- Token classification for NER task
- Performance evaluation of NER models (F1 score, precision, recall, cross-entropy)
- Comparison of results with other existing models and studies
- **Python**

Deep Learning for Image Restoration and Synthesis

MVA

Course Content :

- Deep end-to-end learning and its application to super-resolution
- Inverse problems, variational, statistical, and hybrid methods
- Generative models for texture synthesis
- Variational autoencoders
- Generative adversarial networks

Project :

- Utilizing iterative algorithms such as PnP, ISTA, LADMM, and PGD to solve optimization problems through proximal operators and proximal gradient descent, ensuring convergence.
- Applying denoising, deblurring, and super-resolution techniques using pretrained neural networks and spatial blur modeling, with consideration of Gaussian noise in degraded images.
- Employing convex and non-convex regularizations, learned cost functions from denoisers, and proximity-based methods to restore images, optimizing parameters (gamma, lambda, beta, sigma) for the algorithms.
- Comparatively evaluating optimization methods (PnP-ISTA, PnP-LADMM, Prox-PnP-PGD) in terms of accuracy, speed, and flexibility, measuring the quality of restored images using PSNR, and adapting existing codes for a relaxed version of Prox-PnP-PGD.
- **Python**